Rave

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3D Box Packing Calculator - Find the Perfect Fit

Enter your items' dimensions and instantly get the optimal box size, with a live 3D preview of the arrangement.

✓ 6-rotation packing algorithm ✓ Check your box mode ✓ Empty space & filler gaps

✓ Drag to rearrange in 3D ✓ Paste a list from AI or Excel ✓ Centimeters or inches & pounds ✓ Free forever

Calculate box | Check your box

Your box (in)	8	10	4

BOX PADDING (in)
Added once to the box length, width and height, not to each item — 0

WRAP PER ITEM (in)
Added to every item's dimensions (e.g. bubble wrap) — 0

Add item — Load example

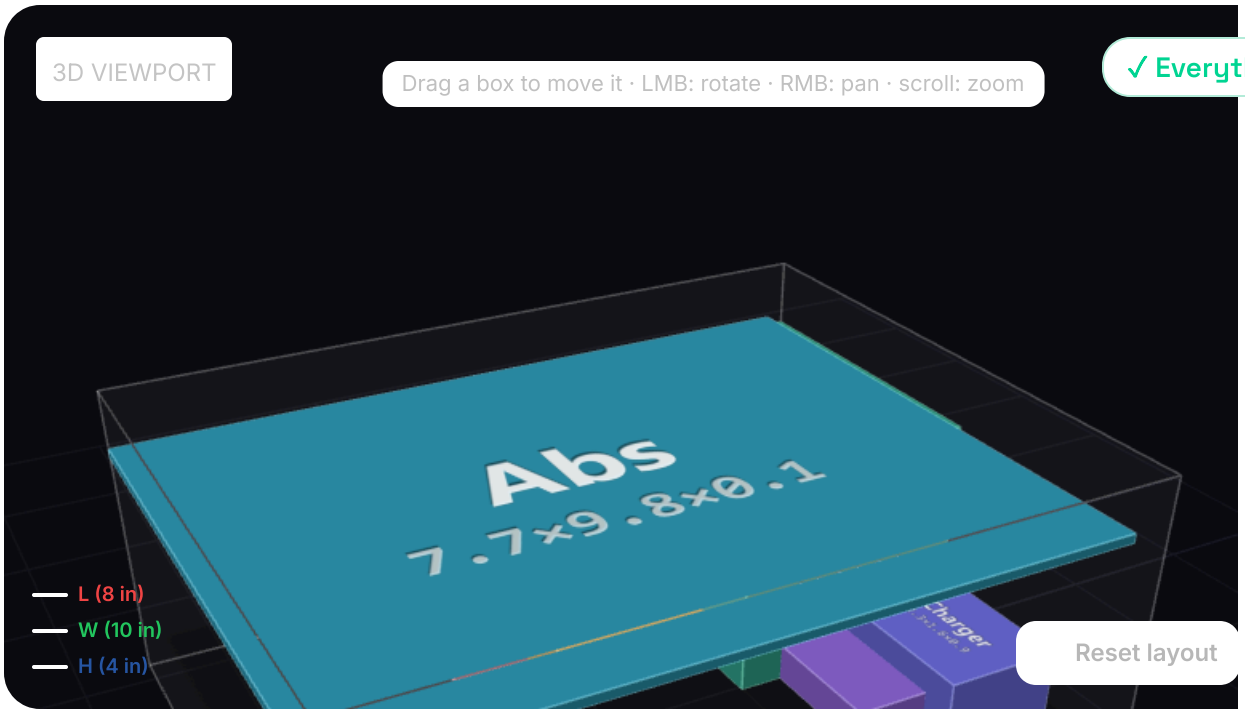
One item | Paste list

Desk top; 47; 24; 1.2; 26; 1
Leg; 27.5; 2.4; 2.4; 2.6; 4
Drawer; 15.7; 13.8; 4.7; 5.5; 2

One item per line: name; L; W; H; weight; quantity (in, lb). Weight and quantity are optional. A unit next to a number (cm, in, kg, lb) wins. Spreadsheet columns, Markdown tables and JSON work too.

Copy prompt for AI

Calculate optimization



ITEM LIST (6/20) Clear all

Charger 2.3×1.6×0.9 in	− 1 + ×	Abs 7.75×9.75×0.1...	− 1 + ×
Battery 4.3×3.2×3.1 in	− 1 + ×	Witty 2.3×1.1×0.4 in	− 1 + ×
Buck 1.8×1×0.5 in	− 1 + ×		
Pi 3.4×2.2×0.67 in	− 1 + ×		

UNITS

cm · kg | in · lb

SHARE RESULT

Copy link

X (Twitter)

Facebook

HISTORY Clear

8×10×4 in Check your box · 6 items	↺ ×
8×10×4 in Check your box · 11 items	↺ ×
8×10×4 in Check your box · 6 items	↺ ×
8×10×4 in Check your box · 6 items	↺ ×

Optimization result

Calculated box

8×10×4 in

Total weight 0.6 lb

Copy size

Empty space

81%

EU 50%

Box padding	0% · 0 in³
Gaps between items	81% · 0.15 ft³
Items	62 in³
Box	0.19 ft³

A smaller box would bring this down to 77%.

Find the smallest box

Gaps for filler

8×6.8×2.4 in	41.3% · 132 in³
8×10×0.8 in	19.5% · 63 in³
3.7×3.2×2.2 in	8.1% · 26 in³
4.6×6.8×0.7 in	6.5% · 21 in³
3.4×3.5×0.7 in	2.5% · 8 in³
1.4×3.2×0.9 in	1.3% · 4 in³
2.3×1.6×0.4 in	0.5% · 1 in³
1.1×1.1×0.7 in	0.3% · 1 in³

2.3×0.6×0.5 in 0.2% · 1 in³

0.5×1×0.5 in 0.1% · 0 in³

Show all gaps in 3D

Every empty spot between items once, largest first, without the box padding. % of the box, so the gaps add up to “Gaps between items”. Slivers thinner than 0.4 in are not listed.

About

Enter the length, width, height and weight of each item, or paste a whole list straight from a spreadsheet or an AI chat, and this calculator runs a **6-rotation bin-packing algorithm** to find the smallest box that fits them all, or checks whether they fit a box you already have. The result renders as an **interactive 3D preview** you can rotate, zoom and rearrange by dragging boxes around: they slide until they touch, lift onto each other when you drag from the side, fall under gravity and swap places when the carton is full. An X-ray view makes every box see-through, so you can check the inside of the stack.

You also see **how much of the box is empty** and why: box padding, wrap around each item, and the real gaps between items, with the exact size of every gap so you know what filler to use. For shipments into the EU, the empty-space figure is shown next to the **50% cap set by the EU packaging regulation (PPWR)** as an estimate, since the official calculation method has not been published yet.

Work in **centimeters and kilograms or inches and pounds**: one switch converts every field, result, gap size and 3D label, while the calculation always uses the exact values you typed. Every calculation runs **instantly in your browser** - no signup required, no uploads. Results can be shared via a link that keeps your arrangement and units, or saved to your local history for later.

Frequently Asked Questions

How does the 3D box packing calculator work? ▾

The 3D box packing calculator uses an extreme-point bin-packing algorithm to find the optimal box size for multiple items. You enter item dimensions, weight, and quantity. The algorithm tests all six rotations of every item at every candidate position and keeps the layout with the smallest outer box after padding, which is also the lowest dimensional weight. The result provides exact box dimensions in centimeters or inches and an interactive 3D preview.

What is the difference between ”Calculate box” and ”Check your box” mode?

"Calculate box" determines the minimum box size required to fit your items. "Check your box" verifies if your items fit inside a specific container you already own. If items do not fit, the calculator displays the exact overflow for length, width, and height. If they fit, it shows how much of the box stays empty and where the gaps are.

How do I measure the empty space in a box?

After you calculate, the Empty space panel shows the share of the box volume not taken up by your items, plus the volume of the items and of the box in liters, or in cubic feet when you work in inches. It works in both modes. Padding and gaps both count as empty space, because that is where filler goes.

How do I know what size filler I need?

The Gaps for filler panel lists the empty blocks left between your items, largest first, with their length, width, and height and their share of the box. Hover or tap a gap to highlight it in the 3D view, or show all of them at once. Every empty spot is listed once, so the gaps add up to the "Gaps between items" figure in the Empty space panel; the box padding is counted separately.

Can I move the items around myself?

Yes. Drag any box in the 3D view to move it. Seen from above, a box slides flat and stops when it touches a neighbor or a wall instead of climbing over it. Seen from the side, you drag it up and across like a wall, so you can lift it onto another box; when you let go, it falls onto whatever is below, and a dashed outline shows where it will land. A box moves in fine 0.5 cm steps and clicks flush against a wall or a neighbor

only when you bring it right up to it, so you can also leave a gap on purpose. Dropping a box onto another one at the same level swaps their places, which is how you rearrange a full carton, and a red outline shows where a box cannot go. Dragging the empty area rotates the view, the right mouse button pans it. A selected box also moves with the arrow keys, 1 cm (or 1 inch) per press, slides until it touches something with Shift + arrow, and rotates with R. The X-ray button makes all boxes see-through so you can check the inside of the stack. The empty-space gaps update after every move, Reset layout returns to the calculated arrangement, and a shared link keeps your arrangement.

What is the EU 50% empty space rule (PPWR)?

The EU Packaging and Packaging Waste Regulation (PPWR, Regulation (EU) 2025/40, Article 24) caps empty space at 50% for grouped, transport, and e-commerce packaging. The cap applies from 1 January 2030, or three years after the European Commission adopts the official calculation method, whichever is later. Filler such as paper, air cushions, or bubble wrap counts as empty space, so adding filler does not lower the ratio; a smaller box does. Until the official method is published, the percentage in this calculator is an estimate, not a compliance check.

How accurate is the box size result?

The packing algorithm produces near-optimal dimensions for shipments of up to 20 item types and 400 pieces. It calculates a physically feasible layout where items never overlap. By default, the calculator adds 3 cm of padding to each dimension for filler material and cardboard thickness. You can adjust this padding to 0 cm to get bare-minimum interior dimensions.

What is the difference between box padding and wrap per item?

Box padding is added once to the length, width, and height of the box, for filler around the whole load and the cardboard walls. The default is 3 cm, so a 3 cm padding makes the box 3 cm longer, wider, and taller in total, not 3 cm per item. In "Check your box" mode it reduces the usable interior instead. Wrap per item is added to every item's dimensions, for example when each item is wrapped in bubble

wrap or foam. You can use both together, such as 3 cm for the box and 1 cm around each item.

Can I paste a whole list of items at once?

Yes. Switch the form to Paste list and paste one item per line as name; length; width; height; weight; quantity, in centimeters and kilograms, or in inches and pounds when the units are set to in · lb. A unit written next to a number (cm, in, kg, lb) always wins. Weight and quantity are optional. Columns copied from a spreadsheet, Markdown tables, LxWxH sizes, and JSON also work. The Copy prompt for AI button gives you a ready instruction for ChatGPT, Claude, or another assistant to turn your product list into this format. Every line is checked, and lines with errors are listed by number instead of being added.

Can I use inches and pounds instead of centimeters and kilograms?

Yes. Set Units under the 3D view to in · lb and every field, result, empty-space and gap figure, label in the 3D view, history entry, copied box size, and shared link switches to inches, pounds, and cubic feet. The calculation always runs on the exact values you typed, so switching units back and forth never changes the result: 12 inches stays 12 inches. Visitors whose device is set to a US time zone start in inches; your own choice is remembered in your browser, and a shared link opens in the units it was shared in.

What is dimensional weight and why does it matter for shipping costs?

Dimensional weight (or volumetric weight) is a pricing metric carriers use to charge for package volume rather than physical weight. Carriers calculate it as Length × Width × Height ÷ 5000 (in cm/kg) and bill the higher of the two values. The calculator picks the layout with the smallest box, which keeps dimensional weight as low as possible.

How many items can I pack at once?

The calculator supports up to 20 distinct item types per calculation. Each item type can have a quantity of 20, allowing up to 400 total pieces per box. Individual items

can measure up to 300 cm (118 in) per side and weigh up to 100 kg (220 lb).

Is my data private when using this calculator?

Yes. Every calculation runs entirely in your browser, and the item names and dimensions you type are not sent to our servers. Your history is stored only in your own browser. The one exception is a share link you choose to create: it carries the items in the URL, so whoever opens it, and the link preview image, can show the same result. Anonymous, cookieless usage counts (for example that a calculation ran) never include what you typed.

Related guides

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EU 50% Empty Space Rule for Shipping Boxes: How to Measure It

The EU packaging regulation (PPWR) caps empty space in e-commerce boxes at 50%. What counts as empty space, how to measure it and how to cut it.

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Cheapest Ways to Ship a Package in 2026 - Carrier Comparison

Compare InPost, DHL, FedEx, UPS, USPS, Royal Mail, and Evri for 2026. Real price tiers, dimensional weight gotchas, and how to pick the cheapest carrier for your parcel.

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